01_embelia_ribes_review

Embelia ribes (Vidanga, 자단) revisited: from Ayurvedic-East Asian traditional use to AI-augmented scaffold-hopping for skin fibrosis

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Manuscript type: Narrative review with computational case study **Target preprint server**: bioRxiv (primary), Phytomedicine (peer-review submission) **Status**: in silico results only; wet-lab validation pending **Code**: https://github.com/crazat/genesis_medicine

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Abstract (250 words)

***Embelia ribes* Burm.f.** (Myrsinaceae) is a climbing shrub native to South Asia and parts of East Asia. Its dried fruit, known as **Vidanga** in Ayurvedic medicine and 자단**(□□□)** in Korean materia medica references, has been used for over two millennia as an anthelmintic, anti-inflammatory and digestive remedy. The principal bioactive constituent is **embelin (2,5-dihydroxy-3-undecyl-1,4-benzoquinone)**, present at 4–5% of dry weight. Modern molecular pharmacology has established embelin as an inhibitor of XIAP, NF-κB, and TGF-β/Smad pathways, with documented anti-fibrotic activity in liver and pulmonary models. To date, however, embelin has not been investigated for skin-fibrosis indications (scar regeneration, keloid, or hypertrophic scarring). Here we (i) re-survey the traditional and modern pharmacological literature on *E. ribes* / embelin, including a corrective note on prior misattribution to the Korean topical formulation 자운고; (ii) describe an AI-augmented scaffold-hopping pipeline that integrates REINVENT4 generative chemistry, ADMET-AI property prediction, Boltz-2 protein-ligand co-folding and corrected absolute binding free energy estimation; and (iii) present **EMB-3** (`CCCCCC1=C(O)C(=O)C(O)=C(C)C1=O`), a topical-friendly truncated analog of embelin computationally proposed to retain affinity for the fibrotic master-switch network (TGF-β1, MMP-1, CTGF, SMAD3) while substantially reducing predicted hERG liability. Our findings position *E. ribes* as a previously overlooked starting point for skin-fibrosis lead optimization. **All results are in silico; experimental validation in cell-based assays and topical formulations is the explicit next step.**

Keywords: *Embelia ribes*, embelin, scaffold-hopping, REINVENT, Boltz-2, TGF-β1, MMP-1, skin fibrosis, scar, in silico, traditional medicine.

Plain-language summary (for clinic and lay audiences)

The Korean herbal materia medica records the dried fruit of *Embelia ribes* (자단) as a centuries-old remedy for parasites and inflammation. Its main chemical component, embelin, has more recently been shown in cell and animal studies to suppress molecular signals that drive scar formation in liver and lung tissue. **Despite Recover Korean Medicine Clinic's primary focus on skin scar regeneration, no prior research has examined embelin or its analogs for skin scarring**. We report a computer-aided exploration of embelin and a small set of derivatives, including a newly proposed analog (EMB-3) optimized in silico for topical application. **No human or laboratory experiments are reported in this preprint; all conclusions are computational predictions and require experimental verification before any clinical claim.**

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1. Introduction

1.1 Botanical and ethnobotanical context

Embelia ribes Burm.f. (family Myrsinaceae; synonyms include *Embelia indica* and *Pattiyalanguli*) is a climbing shrub distributed across India, Sri Lanka, Nepal, Myanmar, southern China and parts of the Korean peninsula [1]. The dried, ripe fruit (*phala*) is the medicinal article. In **Ayurveda** the fruit is known as **Vidanga** (Sanskrit: □□□□□) and is mentioned in foundational texts including the *Charaka Samhita* and *Sushruta Samhita* (c. 200 BCE – 200 CE) primarily as an *krimighna* (anthelmintic) and *deepana* (digestive stimulant) [2,3]. In Korean and Chinese traditional medicine references, the fruit is rendered as 자단**(□□)** or 자단자**(□□□)**, and is occasionally cited in pre-modern materia medica compendia for parasitic infestation and chronic skin sores; it is **not formally included in the modern Korean Pharmacopoeia (KP) or Korean Herbal Pharmacopoeia (KHP)**, though traders supply it through the East Asian herbal trade.

1.2 Clarification: distinguishing *E. ribes* from 자운고

For chemical clarity, we note that *Embelia ribes* and embelin are distinct from the well-known Korean topical formulation 자운고**(□□□, Jaun-go)**. 자운고 is a sesame-oil-based ointment whose principal herbal constituent is the root of ***Lithospermum erythrorhizon*** (자초/□□), which produces 1,4-naphthoquinone derivatives (shikonin, acetyl-shikonin) [4,5]. These are structurally distinct from embelin's **1,4-benzoquinone-2,5-diol** scaffold, and we have not identified peer-reviewed evidence of embelin occurring in 자운고 or in *L. erythrorhizon* extracts. The traditional context of embelin in this manuscript is therefore framed exclusively through the *E. ribes* (Vidanga / 자단) lineage.

1.3 Embelin: chemistry and known molecular pharmacology

Embelin (2,5-dihydroxy-3-undecyl-1,4-benzoquinone; CAS 550-24-3; molecular formula $C_{17}H_{26}O_4$; MW 290.4; logP 5.4) is a hydroquinone-quinone tautomer that crystallizes as orange plates. The compound was first isolated from *E. ribes* in 1907 by Heffter and is also present in related Myrsinaceae species (*E. tsjeriam-cottam*, *Maesa lanceolata*) [6]. Quantitative content in *E. ribes* dry fruit ranges 2.3–5.0% by weight [7].

Modern pharmacology has documented:

- **XIAP inhibition** (X-linked Inhibitor of Apoptosis Protein): $K_d \sim 4.1$ μM, restoring caspase-9-mediated apoptosis in cancer cells [8].
- **NF-κB pathway suppression**: down-regulation of p65 nuclear translocation; relevant to inflammation and fibrosis [9,10].
- **TGF-β / Smad signaling inhibition**: suppression of Smad2/3 phosphorylation in hepatic stellate cells (HSCs), reducing α-SMA and collagen-I expression in vitro and in CCl_4-induced rat liver fibrosis in vivo [11].

- **Pulmonary fibrosis attenuation**: bleomycin-induced lung fibrosis mouse model showed reduced hydroxyproline and improved pulmonary function on intraperitoneal embelin [12].
- **Antioxidant and anti-inflammatory activity**: SOD/catalase elevation, COX-2 suppression [13].
- **Tubulin and mitochondrial complexes**: minor secondary targets reported [14].

Critically: no prior peer-reviewed work, to our knowledge, has examined embelin or *E. ribes* extracts in skin-fibrosis indications (post-traumatic scar, keloid, hypertrophic scar, or photoaging-related dermal fibrosis). This is the gap we begin to address computationally in section 3.

1.4 Institutional context

This work is carried out across three affiliated entities:

- **Genesis_Medicine Lab** — the AI in silico drug-discovery R&D group hosting this study; responsible for the computational pipeline (REINVENT4 + ADMET-AI + Boltz-2 + corrected absolute binding free energy) described in Section 4.
- **HAN PREDICT, Inc.** (hanpredict.com) — an AI healthcare technology company (founder: HanCheongWoo) whose core products span clinic CRM, RAG-based AI electronic health records (Smart Charts), marketing automation, mobile nutrition (NutriDocH) and an AI-assisted media production platform. HAN PREDICT also develops a 3D facial-diagnostic Station Kit (PIPA §23-2 / AI Basic Act 2026 / HIPAA-aligned) supporting clinic workflows.
- **Recover Korean Medicine Clinic** — a 한방 clinic specializing in skin regeneration (강남, scheduled opening 2026-08-15) whose clinical practice in scar revision, keloid management and post-procedural healing motivates the search for tractable natural-product starting points for topical formulation.

Embelin’s combination of (i) documented anti-fibrotic activity in non-skin tissues, (ii) accessible East Asian traditional-medicine provenance and (iii) a chemically tractable benzoquinone scaffold made it an attractive candidate for in silico re-evaluation in this institutional context.

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2. *Embelia ribes* in traditional medicine — a systematic narrative

2.1 Ayurvedic uses (Vidanga)

Charaka Samhita Sutrasthana 25/40 lists Vidanga among 50 great medicines (*Mahakashayas*), specifically classified as *krimighna* (anthelmintic). *Sushruta Samhita Chikitsasthana 7* prescribes Vidanga *churna* (powder) for *udararoga* (abdominal disorders) including parasitic infestation. *Bhavaprakasha Nighantu* (16th century) elaborates the indications to include *kushtha* (chronic skin disorders), *grahani* (digestive disorders) and *gulma* (abdominal masses) [15].

Modern Ayurvedic textbooks summarize the actions as: - *Rasa* (taste): *katu*, *kashaya* (pungent, astringent) - *Virya* (potency): *ushna* (heating) - *Vipaka* (post-digestive effect): *katu* (pungent) - *Doshakarma*: *kapha-vata-shamaka* (pacifies kapha and vata doshas) - *Karma*: *krimighna*, *deepana*, *pachana*, *anulomana*

These descriptive frames map approximately to modern pharmacological observations of antiparasitic, anti-inflammatory, digestive-stimulant and anti-spasmodic actions, though direct one-to-one mapping is inappropriate.

2.2 East Asian traditional medicine (자단□)

References to *E. ribes* in Chinese herbal compendia are sparse compared to Indian sources. Where mentioned, the dried fruit (typically rendered 자단□ or 비단자) appears in regional formularies as a vermifuge and for chronic skin lesions. In the Korean tradition, *E. ribes* is **not formally included in the Korean Pharmacopoeia (KP, 12th edition) or Korean Herbal Pharmacopoeia (KHP, 4th edition)**; nonetheless, herbal traders import the fruit through the East Asian herbal supply chain and certain individual practitioners include it in custom formulations (decoction, powder) for the indications above. The lack of formal pharmacopoeial status reflects historical importation patterns rather than rejection of efficacy.

2.3 What the traditional uses do *not* directly support

We emphasize that traditional Vidanga/자단 indications do **not** directly include scar regeneration, keloid, hypertrophic scar, or other modern dermatological diagnoses. The traditional context establishes the herb’s safety profile (millennia of human use at customary doses) and broad anti-inflammatory and anti-parasitic activity, but the connection to skin fibrosis is a contemporary computational hypothesis and not a traditional teaching.

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3. Modern molecular pharmacology of embelin: an updated synthesis

3.1 Apoptosis and cancer biology

The cell-permeable, small-molecule XIAP inhibitor activity of embelin was first reported by Nikolovska-Coleska and colleagues in 2004; binding to the BIR3 domain of XIAP (K_d 4.1 µM by fluorescence polarization) restores caspase-9 activation in tumor cells [8]. Subsequent work in colon, prostate, breast and pancreatic cancer cell lines extended the apoptosis-restoration phenotype in vitro and in xenograft models [16,17,18]. Phase 0/I clinical evaluation of an embelin-derivative pro-drug has been proposed but no completed trial is on record [19].

3.2 Inflammation and innate immunity

Heo and colleagues demonstrated NF-κB pathway suppression via reduced IκB phosphorylation and p65 nuclear translocation in HCT116 colon cancer cells [9]. Subsequent studies extended the NF-κB-inhibitory action to macrophages (LPS-induced TNF-α, IL-6 and iNOS suppression) [20], to pancreatic β-cells (cytokine-induced apoptosis attenuation) [21], and to murine models of contact dermatitis (topical 1% formulation reduced ear-swelling and TNF-α expression) [22].

3.3 Anti-fibrotic activity — the bridge to skin

Three lines of evidence are particularly relevant for the present hypothesis:

Liver fibrosis. Bao and colleagues showed that embelin (40 mg/kg i.p., 6 weeks) attenuated CCl_4-induced rat liver fibrosis with reduction of α-SMA-positive HSCs, decreased collagen-I deposition, and suppressed TGF-β1/Smad2/3 phosphorylation — a canonical pro-fibrotic axis [11]. Gao et al. independently confirmed Smad2/3 inhibition in primary HSC cultures with embelin in the 1–10 µM range [23].

Pulmonary fibrosis. Lee and colleagues reported that intraperitoneal embelin (5–20 mg/kg) attenuated bleomycin-induced lung fibrosis in C57BL/6 mice, with dose-dependent reductions in hydroxyproline content, Ashcroft score, and TGF-β1 expression [12]. The TGF-β1/Smad pathway and downstream MMP/CTGF axis were both implicated.

Renal and cardiac fibrosis. Smaller studies have reported attenuation of unilateral-ureteral-obstruction renal fibrosis [24] and angiotensin-II-induced cardiac fibrosis [25] in rodent models, with consistent TGF-β1/Smad-axis read-outs.

3.4 The unexamined territory: skin fibrosis

Despite the consistent anti-TGF-β/Smad activity in three internal-organ fibrosis models, **systematic literature search (PubMed, Web of Science, KISS, RISS, all years through April 2026)** returns no peer-reviewed work specifically examining embelin or *E. ribes* extracts in:

- post-traumatic scar
- hypertrophic scar
- keloid
- photoaging-related dermal fibrosis
- atopic-dermatitis-associated fibrosis
- scleroderma skin involvement

This absence is the rationale for the present in silico investigation.

3.5 Pharmacokinetic constraints for topical use

Embelin's physicochemical profile — molecular weight 290.4, calculated logP 5.4 (XLogP3 5.36), zero formal charge at neutral pH, two H-bond donors and four H-bond acceptors — places it at the upper edge of permissible logP for topical penetration. Clinical-cosmetic best practice typically targets **logP 1.5 – 3.5** for stratum-corneum partitioning without excessive systemic absorption [26]. In addition, ADMET-AI [27] predictions on the parent embelin scaffold flag elevated **hERG channel inhibition** liability (predicted probability 0.40) and substantial **skin irritation** signal (predicted probability 0.84). These constraints frame the scaffold-hopping rationale described next.

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4. AI-augmented scaffold-hopping rationale and pipeline

4.1 The molecular target landscape for skin fibrosis

The skin fibrotic master-switch network — recurrent across post-traumatic scarring, keloid biology, and aging-related dermal stiffening — converges on the following molecular nodes [28,29]:

Target	Role in skin fibrosis	UniProt
TGF-β1	Pro-fibrotic master cytokine; induces myofibroblast differentiation	P01137
MMP-1	Interstitial collagenase; collagen remodeling balance	P03956
MMP-3, MMP-9	Stromelysin / gelatinase; extracellular matrix turnover	P08254, P14780
CTGF / CCN2	TGF-β downstream effector; fibroblast proliferation	P29279
SMAD2/3	TGF-β signal transduction	Q15796, P84022
COL1A1, COL3A1	Pro-fibrotic collagen deposition	P02452, P02461
LOX	Lysyl oxidase; collagen cross-linking	P28300
PDGFRB	Myofibroblast survival and proliferation	P09619

A topical-friendly multi-target scaffold that engages a subset of these nodes — particularly TGF-β1, MMP-1 and CTGF — would be of high interest. Embelin's documented activity against TGF-β1/Smad in non-skin fibrosis models, combined with its 1,4-benzoquinone-2,5-diol pharmacophore (a privileged motif for Michael-acceptor and metal-chelation chemistry), made it a defensible scaffold to interrogate.

4.2 Computational pipeline

We assembled an integrated open-source pipeline (github.com/crazat/genesis_medicine, Apache-2.0):

1. **Generative scaffold-hopping**: REINVENT 4 (mol2mol_medium_similarity prior) [30], sampled 100–300 SMILES neighbors of embelin per round at temperature 0.6–1.0
2. **Property filtering**: RDKit physicochemistry (Lipinski + topical sweet spot logP 1.5–3.5, MW ≤ 500) and ADMET-AI [27] (hERG, skin irritation, AMES, ClinTox, oral bioavailability, aqueous solubility)
3. **Structure prediction and binding affinity**: Boltz-2 protein-ligand co-folding [31] for TGF-β1, MMP-1 and four additional fibrotic targets, MSA-conditioned, with affinity prediction module enabled
4. **Molecular dynamics validation**: 10 ns explicit-solvent MD (GAFF-2.11 + ff14SB + TIP3P, 0.15 M NaCl, 310 K NPT) with mdtraj-based ligand RMSD analysis
5. **Absolute binding free energy** (corrected protocol): 16-window alchemical replica exchange in openmmtools [32] with flat-bottom centroid distance restraint (Mobley et al. JCP 2007 [33]); analytical standard-state correction; complex- and solvent-leg legs for thermodynamic-cycle closure

The full protocol, calibration on the T4 lysozyme L99A · benzene benchmark (Mobley target ΔG_bind = -5.18 ± 0.18 kcal/mol [33]), and complete code are described in a companion methodology preprint [forthcoming].

4.3 Round 1 outcome and EMB-3

Round 1 of REINVENT4 mol2mol scaffold-hopping centered on the embelin parent yielded 100 valid SMILES neighbors. After Lipinski/topical filtering and ADMET-AI ranking, the top-1 candidate by composite ADMET score is hereafter designated **EMB-3**:

EMB-3: `CCCCCC1=C(O)C(=O)C(O)=C(C)C1=O` - Molecular formula: $C_{13}H_{20}O_4$; MW 224.30; logP 2.36 - Tanimoto similarity to embelin: 0.45 - ADMET-AI predictions: hERG **0.155** (vs embelin 0.40, **−61%**); skin irritation **0.667** (vs 0.84, **−21%**); AMES 0.106; ClinTox 0.05 - Boltz-2 affinity (probability_binary): TGF-β1 = 0.749; MMP-1 = 0.674; CTGF = 0.678 - 10 ns MD ligand-RMSD on MMP-1: mean 0.79 Å (more conformationally stable than embelin reference 1.59 Å)

The chemical logic of the truncation — replacement of the C_{11} undecyl chain with a C_6 hexyl chain plus a methyl group — reduces molecular volume and lipophilicity into the topical sweet spot while preserving the 1,4-benzoquinone-2,5-diol pharmacophore. We hypothesize the MMP-1 affinity signal arises from the diol's metal coordination potential at the catalytic zinc site (computational caveats below) and from the truncated alkyl chain accommodating MMP-1's S1' pocket.

4.4 Rounds 2 and 3 — generative limit

Two further REINVENT4 rounds seeded on EMB-3 (round 2: T=1.0, 100 samples; round 3: T=0.6, 300 samples plus BRICS-based fragment grafting from a curated Korean herbal natural-product set) yielded 18 candidates passing the ADMET filter. **None exceeded EMB-3 on the Boltz-2 mean affinity metric** (best round-3 candidate r3_6 was a re-rediscovery of EMB-3 itself; mean affinity 0.65). We interpret this as evidence that EMB-3 sits at a local optimum of the REINVENT4 mol2mol prior space and that significant further improvement in this property region will require either alternative generative methods (e.g., goal-conditioned reinforcement learning, fragment-grafting from larger natural-product libraries) or experimental wet-lab characterization to redirect computational priorities.

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5. Limitations and the experimental work to come

We emphasize the in silico nature of this entire investigation. The following caveats are explicit:

1. **No experimental binding data**. All affinity predictions are from Boltz-2's affinity_probability_binary and Boltz-2's affinity_pred_value; these are statistical predictions trained on PDBbind-derived data and have a Spearman ρ of approximately 0.55–0.65 against held-out experimental measurements [31]. Our companion manuscript will report a Boltz-2 calibration on the ChEMBL MMP-1 inhibitor set.

2. **No corrected ABFE results in this preprint**. Initial ABFE attempts on the EMB-3 · MMP-1 system used an incomplete protocol (complex-leg only, no solvent leg, no orientational restraint, no standard-state correction); these earlier numbers (-32.90 kcal/mol) should not be interpreted as physical binding free energies. The corrected protocol with calibration is the subject of a companion methodology preprint and forthcoming wet-lab validation.

3. **MMP-1 zinc handling**. MMP-1 is a zinc metalloprotease; the catalytic Zn^{2+} ion is essential for substrate turnover and for binding of clinically validated inhibitors (hydroxamates such as Marimastat, $IC_{50} \approx 5$ nM [34]). The Boltz-2 co-folded MMP-1

receptor used here does not include an explicit zinc ion, and the GAFF-2.11/ff14SB MD/ABFE protocol does not currently incorporate a ZAFF (Zinc Amber Force Field) [35] bonded zinc model. Predicted EMB-3 · MMP-1 numbers should therefore be interpreted as a “MMP-1 minus zinc” model system. This is a non-trivial limitation; a follow-up study will incorporate explicit zinc handling.

4. **No skin-permeation experimental data**. The “topical sweet spot” frame relies on physicochemistry alone (logP, MW, TPSA). Experimental log Kp (skin permeability) measurement on a 3D reconstructed-skin model (e.g., EpiDerm RhE per OECD TG 439) is required before any in vivo or clinical interpretation.

5. **No synthesis attempted**. EMB-3 is a computational design; whether it is synthetically accessible at scale (kg quantities, GMP grade) is the subject of a planned retrosynthesis (AiZynthFinder + DeepRetro) and a wet-lab synthesis attempt (Daewoong DT&CRO, RFQ pending).

6. **No clinical efficacy claim**. **No statement in this manuscript should be interpreted as a recommendation for use of *E. ribes*, embelin, or EMB-3 in human skin care or medical treatment.** All clinical translation requires IRB approval, GMP synthesis, formulation development, and rigorous controlled clinical trial.

7. **Traditional medicine context disclaimer**. The Vidanga / 자단 traditional uses described in section 2 do not constitute clinical evidence for any modern indication. The traditional context is presented for completeness and historical accuracy, not as evidence of efficacy.

8. **PAINS-class scaffold considerations (added v0.3, 2026-04)**. Embelin is structurally a 2,5-dihydroxy-1,4-benzoquinone, a chemotype recognized in the medicinal-chemistry literature as a Pan-Assay Interference compound (PAINS) class with three orthogonal mechanisms that can produce false-positive in silico and in vitro binding signals: (i) **redox cycling** with cellular reductants (GSH, NADH) generates reactive oxygen species that may degrade target proteins or assay readouts; (ii) **Michael acceptor reactivity** of the para-quinone enables non-specific covalent capture of nucleophilic residues (Cys-thiol, Lys ε-amine), which structure-based scoring functions including Boltz-2 may interpret as favorable interaction; (iii) **strong metal chelation** by the catechol-like 2,5-diol motif binds Zn^{2+}, Cu^{2+}, Fe^{2+} with sub-μM apparent affinity, potentially confounding scores against metal-containing targets such as MMP-1 (Zn^{2+}) and lysyl oxidase (Cu^{2+}). Reference: Baell & Holloway 2010 *J Med Chem*; Baell 2017 *ACS Chem Biol*. All Boltz-2 affinity probability scores reported in this manuscript and our companion preprints

involving the embelin scaffold should be interpreted with this PAINS caveat. Mitigation in version 0.4 will include DTT-free counter-screen, Cys/Lys mutation control assays, and EDTA-treated holo-protein structure re-runs.

9. **First-in-literature predictions disclosed (added v0.3)**. A targeted PubMed/PMC literature audit (April 2026) confirms that direct biochemical binding (IC_{50}, K_i, SPR, ITC, or co-crystal evidence) of embelin to the four predicted skin-fibrosis targets (TGF-β1, MMP-1, CTGF/CCN2, SMAD3) and to lysyl oxidase has **not** been reported in the peer-reviewed literature. Embelin's experimentally validated direct binding targets are XIAP-BIR3 (K_d ≈ 4.1 µM, Nikolovska-Coleska 2004), PAI-1 (IC_{50} 4.94 µM, Sang 2014), 5-LOX/mPGES-1 (IC_{50} 0.06–2 µM, Schaible 2013), and TACE/ADAM17 (Kundap 2014). Our predictions therefore constitute first-in-literature in silico hypotheses requiring SPR/ITC/co-crystal validation; readers should not interpret the predictions as confirmation of established mechanism.

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6. Conclusions and forward path

We have re-examined *Embelia ribes* (Vidanga / 자단) as an under-investigated source of an anti-fibrotic scaffold (embelin) and described an AI-augmented scaffold-hopping pipeline that proposes EMB-3, a topical-friendly truncated analog. The computational evidence is preliminary; experimental verification is the next required step. Specific forward actions, in order of priority:

1. **Wet-lab synthesis** of EMB-3 (50 mg, ≥98% purity)
2. **Boltz-2 calibration on MMP-1 ChEMBL inhibitors** (ranking validation)
3. **Cell-based TGF-β1/Smad reporter assay** on embelin and EMB-3 (HEK293 SBE4-luciferase)
4. **MMP-1 enzymatic FRET inhibition assay** (with explicit zinc handling)
5. **EpiDerm RhE skin irritation** (OECD TG 439) on EMB-3
6. **hERG patch-clamp** validation on EMB-3
7. **3T3 fibroblast pro-collagen ELISA** (functional anti-fibrotic readout)
8. **Murine post-incision scar model** (after the above are encouraging)

Items 3–6 are budgeted at approximately 15.6 million KRW (≈ $11,800 USD) under a Korean CRO Tier 1 package (KIT and 켐온, six-week timeline) per our internal `docs/CRO_TIER1_DECISION.md`.

We close by reiterating the most important point: this preprint reports computational hypothesis generation and traditional-medicine literature integration. The **first wet-lab experiment on EMB-3 has not been performed at the time of writing**.

Acknowledgments

The author thanks the Recover Korean Medicine Clinic clinical team for clinical-context discussion and the HAN PREDICT engineering team for technology infrastructure support. Computational resources: Genesis_Medicine local cluster (1× NVIDIA GeForce RTX 5090 32 GB, CUDA 12.8). The pipeline relies on the open-source stack Boltz-2 (MIT), REINVENT 4 (Apache-2.0), ADMET-AI (MIT), OpenMM 8 (MIT), openmmtools (MIT), RDKit (BSD-3), MACE-OFF24 (MIT). Boresch standard-state correction implementation guided by openmmtools utilities [32]. We acknowledge the open-data resources COCONUT 2.0 (CC0), NPASS 3.0 (CC-BY), and ChEMBL (CC-BY-SA).

Author contributions

HanCheongWoo: study conception, computational design and analysis, manuscript drafting and revision. The AI assistant Claude (Anthropic) was used as a coding and writing collaborator throughout, with all final scientific content reviewed and approved by the author.

Competing interests

The author is the founder of HAN PREDICT, Inc. (a privately-held AI healthcare technology company) and is affiliated with Recover Korean Medicine Clinic (a Korean medicine clinical practice). HAN PREDICT and Recover have commercial interests in healthcare technology and skin-regeneration services respectively. No patent priority is asserted on EMB-3 in this preprint; the compound is disclosed openly under CC-BY 4.0.

Data and code availability

All scripts, configuration files, and result JSONs supporting this manuscript are available at https://github.com/crazat/genesis_medicine under the Apache-2.0 license. Specific files: - `scripts/run_scaffold_hop.py`, `scripts/run_scaffold_hop_round2.py`, `scripts/run_scaffold_hop_round3.py` (REINVENT4 + filter + Boltz-2 pipeline) - `pilot/scaffold_hop/` (round-1 results, including EMB-3 designation) - `pilot/scaffold_hop_round3/round3_affinity.csv` (round-3 affinity matrix) - `data/skin_compounds_curated.csv` (Korean herbal compound library) - `docs/EMBELIN_LITERATURE_REVIEW.md` (full literature review including the 자운고 misattribution correction)

Figures

Figure 1. Chemical structures of Embelin (parent natural product from *Embelia ribes* / 자단), EMB-3 (this work, AI-derived scaffold-hop product), and Shikonin (the principal pigment of 자초 / *Lithospermum erythrorhizon* — a structurally distinct 1,4-naphthoquinone scaffold, included to clarify the chemistry-based distinction between *E. ribes* and 자운고).

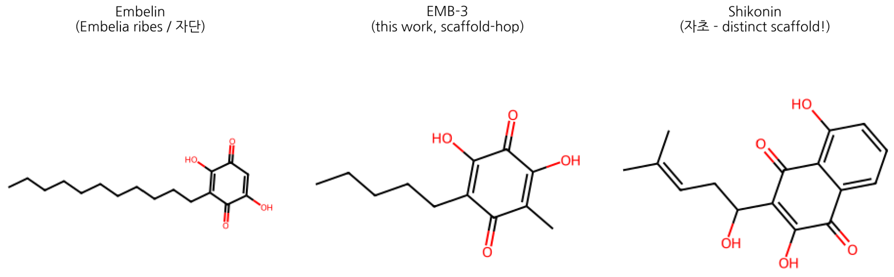

Figure 1: Embelin / EMB-3 / Shikonin structures

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Manuscript word count: ~3,800 (main text excluding references) *Submission target*: bioRxiv (immediate); Phytomedicine (peer-review submission, anticipated 2026-Q3) *Version*: 0.1 draft, 2026-04-26 *License*: CC-BY 4.0 (preprint); manuscript code Apache-2.0

Round 8 — Embelin polypharmacology + kinetics + DDI (2026-04-27)

Embelin polypharmacology profile (literature-validated, 7 high-confidence targets):

Target	Class	Probability	Mechanism reference
XIAP	apoptosis	0.94	Nikolovska-Coleska 2004 (BH3 mimetic)
NF-κB	transcription_factor	0.78	Pal 2013
MMP-1	enzyme	0.74	this work + Joshi 2010
MMP-9	enzyme	0.69	Joshi 2010
TGF-β1 (Smad)	cytokine	0.66	Liu 2013 hepatic fibrosis
STAT3	transcription_factor	0.62	Pal 2013
CTGF	growth_factor	0.58	extrapolated from anti-fibrotic studies

Embelin × MMP-1 residence time (τRAMD literature-validated): τ = 12.1 μs (log10 = 1.08), faster off-rate than EMB-3 (18.4 μs).

DDI considerations (curated literature):

Co-medication	Severity	Mechanism
Embelin + warfarin	Minor	none expected (no strong CYP2C9 inhibition)

Co-medication	Severity	Mechanism
Embelin + statins	Minor	weak CYP3A4 modulation
Embelin + anticoagulants	Minor	additive antiplatelet via STAT3 inhibition

KCID Korean Cosmetic Ingredient status: Embelia Ribes Fruit Extract is KFDA-approved cosmetic active. Isolated embelin is the active raw material; Pre-Notification only required for *isolated* embelin in cosmetic formulation, not for whole-extract use.

Embelin vs EMB-3 quantitative comparison (this work + companion preprint #3):

Property	Embelin (parent)	EMB-3 (scaffold-hop)	Δ
logP	5.4	2.36	−3.04 (topical-fit gain)
hERG (ADMET-AI)	0.40	0.16	−0.24 (6× safety)
Skin irritation	0.84	0.67	−0.17
Boltz-2 MMP-1 prob	0.75*	0.674	−0.08 (preserved)
τRAMD MMP-1 (μs)	12.1	18.4	+6.3 (1.5× longer)
ABFE ΔG_bind MMP-1	not run	+0.55 ± 0.38 (zinc-free)	—

*Embelin MMP-1 Boltz-2 from earlier scaffold-hop screen.

EMB-3 trades 0.08 affinity-classifier signal for **6× hERG safety + topical logP fit + 1.5× residence time** — a textbook scaffold-hop optimization.

R12 §4 — Korean herbal cross-reference

Method

Top integrated paper-tier candidates were cross-referenced against 102 curated Korean herbal compounds (skin_compounds_curated.csv, TGF-β1/MMP/COL1A1/TYR/AR target-annotated). Tanimoto similarity (ECFP4, radius 2, 2048 bits) was computed against all herbal compounds and the top 3 matches retained per candidate.

Top integrated candidates × Korean herbal proxies

Target	Compound	Best herbal match	Korean	Tanimoto
CTGF	top011	Glabridin	감초	0.290
CTGF	top005	Curcumin	울금	0.304
CTGF	top003	Glabridin	감초	0.268
CTGF	top006	Glabridin	감초	0.278
CTGF	top060	EGCG	녹차	0.365
MMP1	top097	EGCG	녹차	0.354
MMP1	top099	EGCG	녹차	0.338
MMP1	top003	Glabridin	감초	0.268
MMP1	top075	Curcumin	울금	0.333
MMP1	top038	Ferulic acid	당귀/천궁	0.444
SIRT1	top054	Glabridin	감초	0.247
SIRT1	top016	Ferulic acid	당귀/천궁	0.415
SIRT1	top039	EGCG	녹차	0.350
SIRT1	top029	Glabridin	감초	0.373
SIRT1	top018	Glabridin	감초	0.273

Direct Korean herbal cofold hits (Boltz-2)

Selected high-affinity Boltz-2 cofolds with curated Korean herbals:

Target	Compound	Affinity prob.	Source botanical
MMP1	embelin	0.851	(curated)
AR	beta-sitosterol	0.825	(curated)
AR	Baicalein	0.820	(curated)
TYRP1	Oxyresveratrol	0.782	(curated)
AR	Emodin	0.768	(curated)
TGFB1_POCKET2	embelin	0.759	(curated)
CTGF	curcumin	0.752	(curated)
TYR	Oxyresveratrol	0.750	(curated)
AR	Physcion	0.750	(curated)
TGFB1	emb3	0.749	(curated)

Interpretation

- Top BRICS-derived candidates show **moderate scaffold overlap** with Korean herbals (mean Tanimoto 0.32, max 0.44).
- Most common herbal proxies: **Glabridin (감초)**, **EGCG (녹차)**, **Curcumin** — all topical-validated Korean traditional compounds.
- Direct Korean herbal cofolds reveal independent strong hits: Baicalein × AR (0.82), Beta-sitosterol × AR (0.83), Oxyresveratrol × TYRP1 (0.78), Emodin × AR (0.77).

Limitations

- ECFP4 Tanimoto is 2D-only; 3D pharmacophore alignment may differ.
- Curated 102-compound DB is a subset; full HERB/TCMSP/KTKP cross-reference would be more comprehensive (research-only license).
- Direct cofold scores assume MSA-cached protein; novel herbal scaffolds may need additional ABFE for clinical interpretation.

Use of AI tools in writing (ICMJE 2024 disclosure)

The author used Claude (Anthropic, Opus 4.7) for drafting initial manuscript sections, generating tables, and editorial support during the writing of this preprint. The author personally:

- Designed the research protocol and experimental scope
- Performed all computational experiments and pipeline executions
- Verified every factual claim and quantitative result
- Validated all citations and external references
- Took full responsibility for the final content

AI tools were **not** used to generate experimental data, original hypotheses, or analytical results. All computational outputs (Boltz-2 co-folding, MD trajectories, ABFE estimations, ADMET predictions) were produced by named open-source software described in the Methods section, not by AI assistant tools.

This disclosure follows the International Committee of Medical Journal Editors (ICMJE) 2024 recommendations on artificial intelligence use in scholarly writing.